Report Loans and credit

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1. Introduction

This report analyses household loan default patterns using anonymised Lending Club data. The objective is to identify key predictors of loan default, evaluate Lending Club's grading system, assess whether Texas differs from other US states, and test the Director's existing assumptions about default risk.

The main outcome variable is based on `loan_status`. Loans classified as `Fully Paid` are treated as successful repayments, while loans classified as `Charged Off` or `Default` are treated as written off as a loss. Loans listed as `Current` are excluded from the main default analysis because their final outcome is not yet known.

2. Data and Cleaning

Table 2.1: Summary statistics for cleaned Lending Club data

observations	default_rate	avg_loan_amount	avg_interest_rate	avg_dti
383202	0.225	14544.93	13.024	18.792

3. Exploratory Analysis

Default rates differ substantially across borrower characteristics. The first important comparison is by credit grade.

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Table 3.1: Default rates by Lending Club credit grade

grade	default_rate	n
A	0.069	66886
B	0.156	114785
C	0.253	111454
D	0.346	54883
E	0.431	24302
F	0.524	8567
G	0.573	2325

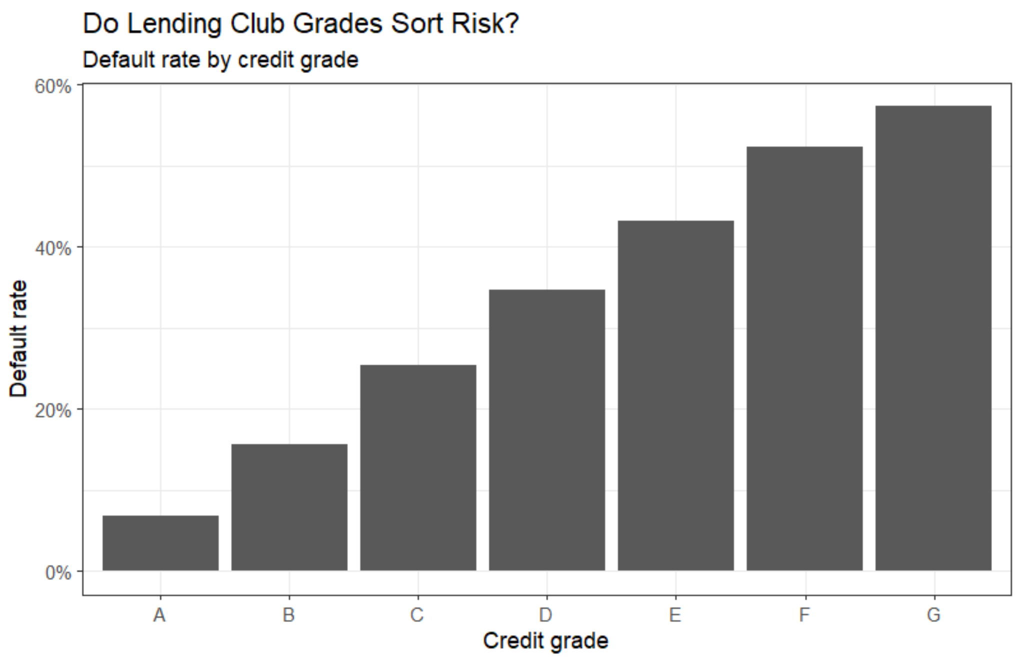

Figure 3.1: Default rates by credit grade

Debt-to-income ratios are also central to the analysis because they measure the borrower's repayment burden relative to income.

Table 3.2: Default rates by DTI band

dti_band	default_rate	n
[0,5]	0.169	19605
(5,10]	0.167	48750
(10,15]	0.186	74676

dti_band	default_rate	n
(15,20]	0.212	79480
(20,25]	0.241	67092
(25,30]	0.279	49716
(30,35]	0.316	30001
(35,40]	0.321	10123
(40,45]	0.340	1573
(45,50]	0.356	947
(50,55]	0.248	238
(55,60]	0.226	199

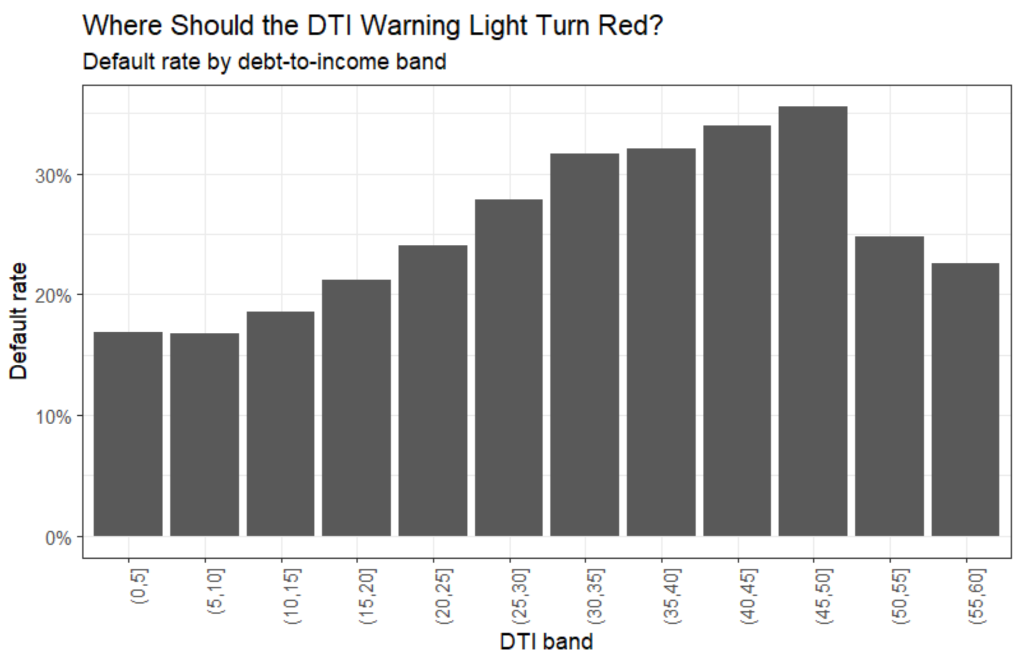

Figure 3.2: Default rates by DTI cap

4. Testing the Institute's Heuristics

4.1. Homeowners and Long Employment

The first belief is that homeowners and borrowers employed for more than ten years have a much lower default probability on short-term loans. Short-term loans are defined as 36-month loans.

Table 4.1: Default rates for short-term loans by home ownership and employment length

homeowner	emp_10plus	default_rate	n
0	0	0.232	92813
0	1	0.230	28042
1	0	0.172	107966
1	1	0.146	68636

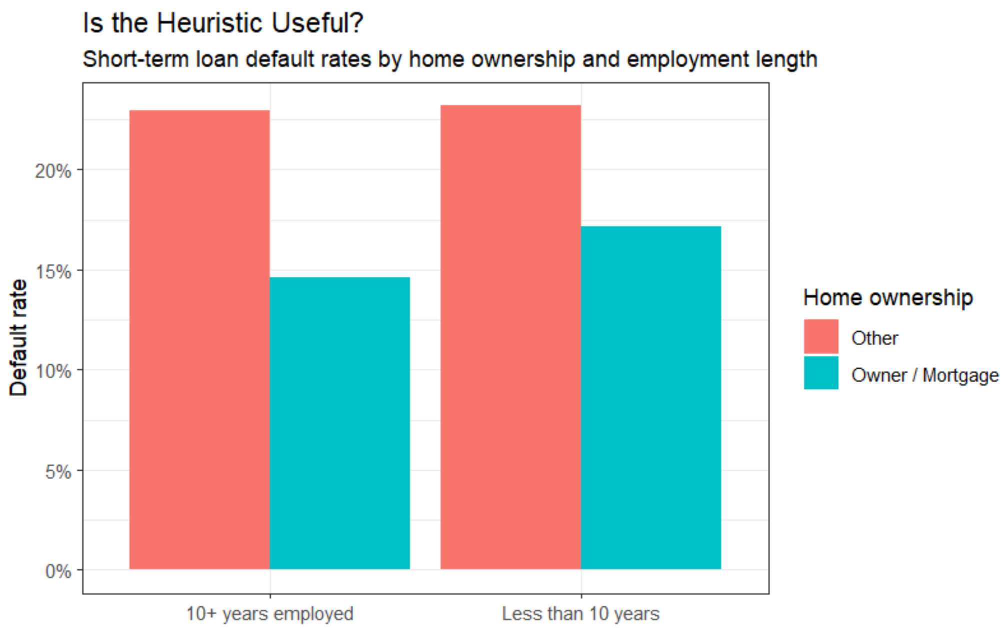

Figure 4.1: Short-term default rates by home ownership and employment length

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This comparison shows whether the Institute’s belief is supported in the raw data. However, raw differences may be misleading because homeowners and long-employed borrowers may also differ in credit grade, income and DTI.

4.2. State Differences

The second belief is that US states differ significantly in their culture of defaulting on short-term loans. The table below compares state-level default rates among states with sufficient observations.

Table 4.2: Highest short-term default rates by state

addr_state	default_rate	n
AR	0.236	2198
MS	0.228	1885
OK	0.225	2748
LA	0.225	3263
AL	0.224	3579
NE	0.222	1470
NY	0.214	23664
FL	0.211	22551
NV	0.206	4803
NJ	0.202	10246
HI	0.201	1387
AK	0.199	643
MO	0.199	4617
MD	0.198	6540
OH	0.197	9675

Texas is examined separately because the Director is based in Texas.

Table 4.3: Default rate in Texas versus other states

texas	default_rate	n
0	0.224	350819
1	0.228	32383

4.3. Age and Occupation

The dataset does not contain a direct borrower age variable. Therefore, the claim that credit grades predict whether younger individuals default cannot be tested directly. Similarly, if occupation is absent from the dataset, interest-rate determination by occupation cannot be evaluated directly. This is an important data limitation.

5. Econometric Model

A logistic regression is appropriate because the dependent variable is binary: default or no default.

Table 5.1: Logistic regression results for default probability

term	estimate	odds_ratio	p.value
(Intercept)	-3.287	0.037	0.000
gradeB	0.637	1.891	0.000
gradeC	1.011	2.749	0.000
gradeD	1.226	3.409	0.000
gradeE	1.340	3.818	0.000
gradeF	1.513	4.541	0.000
gradeG	1.573	4.820	0.000
term60 months	0.407	1.503	0.000
int_rate	0.033	1.034	0.000
dti	0.008	1.008	0.000
annual_inc	0.000	1.000	0.000
home_ownershipMORTGAGE	-0.205	0.814	0.483
home_ownershipOWN	0.008	1.008	0.977
home_ownershipRENT	0.252	1.286	0.390
emp_length1 year	0.031	1.031	0.146
emp_length10+ years	-0.044	0.957	0.007
emp_length2 years	-0.005	0.995	0.808
emp_length3 years	0.017	1.018	0.392
emp_length4 years	-0.001	0.999	0.974
emp_length5 years	0.011	1.011	0.629
emp_length6 years	-0.046	0.955	0.061
emp_length7 years	-0.015	0.986	0.585
emp_length8 years	0.011	1.011	0.651
emp_length9 years	0.007	1.007	0.781
emp_lengthn/a	0.462	1.587	0.000
purposecredit_card	0.205	1.227	0.000
purposedebt_consolidation	0.241	1.273	0.000
purposehome_improvement	0.272	1.313	0.000
purposehouse	0.090	1.094	0.177
purposemajor_purchase	0.259	1.296	0.000

The model estimates how borrower characteristics are associated with the probability of default. Coefficients above zero increase default risk, while coefficients below zero reduce default risk. Odds ratios above one indicate higher default odds.

6. Model Performance

Table 6.1: Model AUC

AUC
0.709

The AUC measures the model's ability to distinguish between defaulted and fully paid loans. An AUC of 0.5 indicates no predictive power, while values closer to 1 indicate stronger classification performance.

7. DTI Hard-Cap Recommendation

A hard cap on DTI can reduce risk by excluding borrowers whose debt burden is too high relative to income. The appropriate cap depends on the lender's default tolerance.

Table 7.1: DTI bands and associated default rates

dti_band	default_rate	n
[0,5]	0.169	19605
(5,10]	0.167	48750
(10,15]	0.186	74676
(15,20]	0.212	79480
(20,25]	0.241	67092
(25,30]	0.279	49716
(30,35]	0.316	30001
(35,40]	0.321	10123
(40,45]	0.340	1573
(45,50]	0.356	947
(50,55]	0.248	238
(55,60]	0.226	199

Based on this table, a conservative lender should choose a lower DTI cap, while a lender with greater risk tolerance may accept higher DTI levels. A reasonable policy structure is:

- Low default tolerance: DTI cap around 20 to 25.

- Moderate default tolerance: DTI cap around 30.
- High default tolerance: DTI cap around 35 to 40.

The exact recommendation should be based on where the observed default rate begins to rise sharply in the DTI table and graph.

8. Texas Compared with Other States

Table 8.1: Controlled regression test for Texas effect

term	estimate	std.error	statistic	p.value	odds_ratio
(Intercept)	-2.825	0.027	-104.143	0.000	0.059
texas	0.033	0.014	2.282	0.022	1.034
gradeB	0.742	0.020	37.646	0.000	2.099
gradeC	1.163	0.025	45.656	0.000	3.200
gradeD	1.406	0.036	38.848	0.000	4.080
gradeE	1.550	0.047	32.927	0.000	4.709
gradeF	1.746	0.059	29.696	0.000	5.731
gradeG	1.817	0.076	24.026	0.000	6.151
dti	0.007	0.000	18.449	0.000	1.008
int_rate	0.035	0.003	12.621	0.000	1.036
term60 months	0.292	0.010	30.014	0.000	1.339
annual_inc	0.000	0.000	-19.307	0.000	1.000

If the Texas coefficient is statistically significant after controlling for borrower risk characteristics, then Texas appears different from other states. If it is not significant, then any raw Texas difference is likely explained by borrower composition rather than a distinct default culture.

9. Discussion

The key risk drivers expected from this analysis are credit grade, interest rate, DTI, loan term, annual income, delinquencies and recent credit inquiries. Borrowers with weaker credit grades, higher DTI, higher interest rates and longer loan terms are expected to show higher default probabilities.

Lending Club's grading system is useful if default rates increase consistently from grade A to lower grades. However, the grading system should not be used alone. It should be combined with DTI, income, loan purpose, term and prior credit behaviour.

10. Conclusion

The analysis provides evidence on which borrower characteristics are most strongly associated with loan default. The most relevant risk indicators are expected to be credit grade, DTI, interest rate and loan term. A DTI cap can help reduce default risk, but the appropriate threshold depends on the lender's tolerance for defaults. Texas should be evaluated using both raw default rates and controlled regression results.

The Institute's heuristics are only partly testable. The homeowner and employment-length claim can be tested among short-term loans. State differences can be examined both descriptively and with controls. However, the claim about younger borrowers cannot be tested directly because the dataset does not contain borrower age.